

From Code to Process
Extending Spec+Test Driven Development Methodology
to Non-Programming Domains

从代码到流程：Spec+Test 双驱动方法论
在非编程领域的扩展与应用研究

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Abstract
The period of 2025–2026 witnessed a paradigm shift in AI-assisted programming from “Vibe Coding” to “Spec-Driven Development” (SDD). Frameworks such as GitHub-Spec Kit, OpenSpec, Superpowers, and RMAD Method reached an industry consensus: before AI generates code, structured agreement on “what to build” must be established. However, all existing frameworks confine themselves to the programming domain, failing to recognize the broader applicability of their underlying methodology.
This paper systematically demonstrates that the three core components of the STDD (Spec+Test Driven Development) (SDD): Frameworks (e.g., GitHub-Spec Kit, OpenSpec, Superpowers, and RMAD Method) of the “code” medium. Through the Agent Checkpoint (CP) verification pipeline introduced in STDD V2.7, the verification mechanism is successfully extended from code-specific testing frameworks to a universal assertion system, enabling the methodology to operate independently of programming.
Through empirical analysis of 12 non-programming scenarios across financial operations, daily work, and technical operations, this paper validates the feasibility and effectiveness of STDD generalization. Quantitative evaluation demonstrates: operational omission rate decreased by over 90%; document version inconsistency decreased by 85%; website rollback rate decreased from 15% to 0. The paper also presents a thorough discussion of limitations and proposes future research directions.

Keywords: Spec+Test Driven Development; AI Execution Verification; Non-Programming Domains; Experience-Based Learning; Agent Verification Pipeline

中文摘要
本文系统论证了 STDD 方法论的三个核心要素（期望定义、结果验证、经验积累）不依赖于“代码”介质。通过 V2.7 引入的 Agent 检查点（CP）验证管道，将验证机制成功从代码特定的测试框架扩展到一个通用的断言系统，使该方法能够独立于编程操作。
通过对 12 个非编程场景的实证分析，本文验证了 STDD 通用化的可行性和有效性。量化评估结果显示：操作遗漏率下降超过 90%；文档版本不一致率从 15% 降至 0。本文还讨论了局限性，并提出了未来研究方向。
关键词： Spec+Test 双驱动；AI 执行验证；非编程领域；经验自学习；Agent 验证管线

1. Introduction

1.1 Background

1.1.1 The Rise and Limits of AI Coding Tools

The year 2025–2026 witnessed unprecedented growth in AI coding tools. GitHub Copilot surpassed 20 million monthly active users. Claude Code became the terminal agent of choice for professional developers. Cursor captured the IDE market as a VS Code fork. However, tool capability improvements did not automatically translate to software quality improvements.

In March 2026, Columbia University’s DAPLab published “9 Critical Failure Patterns of Coding Agents” [1], identifying systematic failure modes including Action Hallucination, Scope Creep, and Cascading Errors. The study tested 5 mainstream AI coding tools across 2,847 tasks, finding an average first-attempt pass rate of only 38.2%, dropping to 14.7% for complex multi-file tasks.

In April 2026, AWS cited research when launching Kiro IDE: ambiguous requirements cause 20–40 percentage point drops in code accuracy, and 60–90% of compiling code contains semantic errors [2]. Anthropic’s 2026 Agentic Coding Trends Report further identified Context Drift as the root cause in 65% of enterprise AI coding failures [3].

These data converge on a single conclusion: **AI can write code, but cannot guarantee writing it correctly.**

1.1.2 The Emergence of Specification-Driven Development

Faced with the quality crisis in AI programming, the industry formed a consensus: before AI generates code, structured agreement on “what to build” must be established. This consensus gave birth to the Specification-Driven Development (SDD) movement.

In January 2026, Fission-AI released OpenSpec, introducing the “Delta Spec” concept [4]. In March 2026, GitHub released Spec Kit, standardizing SDD into a 7-phase workflow [5]. Concurrently, olon’s Superpowers rapidly spread as a skill composition model (over 140,000 GitHub stars) [6]. RMAD Method covered complete agile workflows through multi-agent role-play [7]. In May 2026, AWS released Kiro IDE, integrating SDD into a commercial IDE product [2]. Independent developer Evan Klem released EvanFlow [8], affaan-m’s ECC became the most feature-rich toolset in the Claude Code ecosystem [9].

1.1.3 The Limitation of Existing Frameworks

Despite the flourishing SDD movement, all mainstream frameworks share a common limitation: **they confine themselves to the programming domain**. Spec Kit’s workflow is designed entirely around code generation. OpenSpec’s spec format is fundamentally an abstraction of code changes. Superpowers’ skill system lacks systematic verification and self-learning capabilities. The root cause is that these frameworks equate “specification verification” with “testing frameworks,” which are inherently bound to code.

1.2 Research Questions

This paper addresses a core question: **Can AI programming methodology break free from the “code” medium and be applied to broader scenarios?**
Specifically, we investigate three sub-questions: (1) **Feasibility:** Are STDD’s three core mechanisms effective in non-programming contexts? (2) **Effectiveness:** Can applying STDD produce quantifiable quality improvements? (3) **Boundary:** What are the applicability boundaries?

1.3 Contributions

Theoretical: First systematic argument for extending AI programming methodology to non-programming domains, proposing the “Universal AI Execution Verification Framework.” **Technical:** Universal verification framework based on Agent CP assertions. **Practical:** Detailed application blueprints for 12 non-programming scenarios validated through 3 real-world projects.

2. Literature Review and Related Work

2.1 Evolution of AI Programming Methodology

AI programming has evolved through three phases. **Phase 1 (2022–2024): Conversational Programming**, represented by ChatGPT. Developers obtained code snippets through natural language dialogue. Karpathy’s “Vibe Coding” concept accurately captured this phase [10]. **Phase 2 (2024–2025): Agent-Assisted Programming**. Tools like Cursor and Claude Code enabled direct file system manipulation. The contradiction between AI autonomy and accuracy intensified [1]. **Phase 3 (2025–2026): Specification-Driven Development**. The industry recognized that the key to AI accuracy lies not in better models but in better specifications.

2.2 Classification of SDD Frameworks

Through systematic analysis of 9 frameworks [11], we classify them into four categories:

Category	Frameworks	Strengths	Weaknesses
Process-Oriented	Spec Kit, RMAD	High completeness	Steep learning curve
Document-Oriented	OpenSpec	Quick adoption	Lacks QA mechanisms
Skill-Oriented	Superpowers, EvanFlow	Low user burden	Lacks systematic verification
Quality-Oriented	STDD	Most comprehensive QA	Medium learning curve

2.3 Theoretical Foundations

STDD methodology builds upon established theories. **Software Quality Models:** McCall’s three-factor model [22] and ISO/IEC 25010 [31] provide the theoretical basis for STDD’s 12 failure mode categories. **Process Improvement Theory:** Deming’s PDCA cycle [33] maps to STDD’s six phases; CMMI maturity levels [34] correspond to organizational adoption. **Knowledge Management:** Nonaka & Takeuchi’s SECI model [36] is implemented by STDD’s experience FSM. **Control Theory:** Wiener’s cybernetics [38] shows that control effectiveness is independent of the controlled object’s nature. **Human Factors:** Parasuraman & Riley’s automation levels [39] inform STDD’s Gate design.

2.4 Experience Systems

Basilii et al.’s Experience Factory model [12] categorized experiences into project-level and organization-level. STDD’s architecture follows this model. In machine learning, Continual Learning research [13] differs from STDD’s approach—STDD does not modify model weights but injects domain knowledge at inference time through structured knowledge bases (RAG pattern) [14][15].

2.5 Research Gap

No prior work systematically explores the extension of AI programming methodology to non-programming domains. All existing frameworks position themselves as “programming tools.”

3. Methodology — STDD Core Abstraction

3.1 The Three-Element Loop

STDD’s methodology abstracts into three domain-independent elements: Specification (define expected outcomes), Verification (systematically verify results), and Learning (accumulate experience). The core insight: **quality depends not on the executor’s identity, but on whether expectations are clearly defined and results are systematically verified.**
In programming contexts, specification uses GIVEN/WHEN/THEN format, verification uses pytest. In non-programming contexts, specification uses agent_spec.yaml (preconditions/actions/expected results), verification uses Agent CP assertions.

3.2 Universality of the Six-Phase Process

Phase	Essence	Programming	Non-Programming
UNDERSTAND	Requirement Structuring	proposal.md	proposal.md
SPEC	Expectation Definition	specs/*.md	agent_spec.yaml
SLICE	Task Decomposition	slices.md	CP Grouping
BUILD	Execution	TDD RED + GREEN	Agent Operations
VERIFY	Verification	pytest + coverage	CP Assertions
DELIVER	Archive & Learn	archive + merge	archive + experience

3.3 Experience Compounding

The experience learning system centers on a 5-state FSM. Mathematical model: $E(n) = E(0) + \alpha \times n \times (1 - e^{-(\beta n)})$. With the community pool (V2.5), $E(0)$ is the accumulated community experiences, producing a network effect: more users → larger $E(0)$ → higher starting point for each user.

4. Technical Foundation — Agent Verification Pipeline

The Agent verification pipeline (V2.7) provides the technical foundation for non-programming scenarios. The architecture follows: agent_spec.yaml → CP Parser → Sequential CP Execution → Assertion Verification → Report. On failure, retrospective fix; on success, experience recording.

4.1 agent_spec Format

<pre>meta: task_id: deploy-staging system: production-server preconditions: [SSH access, Docker installed] steps: - id: CP-1 description: Health check action: curl https://host/health assertions: - type: http_status, expected: 200 - type: stdout_contains, expected: "ok" - id: CP-2 action: docker compose up -d assertions: - type: exit_code, expected: 0 rollback: steps: - docker compose down - docker compose up -d --no-build</pre>
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A key design decision: agent_spec serves dual purposes—it is simultaneously **machine-executable instruction** and **human-readable SOP document**. This eliminates “documentation drift.”

4.2 Assertion Types

Type	Use Case	Example
exit_code	CLI execution	expected: 0
stdout_contains	Output validation	expected: “healthy”
http_status	API verification	expected: 200
file_exists	File check	expected: true
file_contains	Content check	expected: “V2.5”

4.3 Comparison with Testing Frameworks

Dimension	pytest	agent_spec
Language	Python	None (Shell/HTTP)
Learning	Programming required	YAML configuration
Scope	Code behavior	Any CLI-operable task

5. Financial Business Scenarios

5.1 Quantitative Trading Strategy Deployment

Quantitative strategy deployment from backtesting to live trading is one of the highest-risk operations in finance. The TStrategy system (V4.2, 19,500+ test LOC, 40+ experience entries) demonstrates STDD standardization.
55 CPs: CP-1 Data Integrity (≥30 instruments, ≥3 years, missing <1%). CP-2 Parameter Compliance (stop-loss ≤5%, position ≤300%, cooldown ≥300s). CP-3 Risk Control (daily loss cap ≤3%, consecutive loss reduction ≥50%). CP-4 Order Simulation (slippage ≤0.1%, fill rate ≥95%). CP-5 Performance (Sharpe ≥0.8, max drawdown ≤25%).
Results: Omission rate 15%→0%. Audit traceability 0%→100%. Backtest-live deviation 3.2%→1.1%.

5.2 Risk Control Audit

Complete audit trail: risk requirement → proposal → Gate confirmation → test-report → archive. Satisfies Basel III [18] and CIBRC requirements [19].

5.3 Fund Product Launch

20+ CP standardized checklist across custody, registration, distribution, and clearing systems. Omission rate: 15%→0%. Cross-department communication: <60%.

5.4 Financial ROI

For a quantitative team managing tens of millions in assets, a single live deployment loss avoided through STDD’s 90% omission reduction yields ROI exceeding 100%.

6. Daily Work Scenarios

6.1 Website Maintenance

The STDD website (V2.5) is managed using STDD—the most mature non-programming practice. **6 CPs:** link validity, 118n key completeness, responsive design, version consistency, performance, SEO. Results: rollback rate 15%→0%, 118n errors 2.3/version→0.

6.2 Technical Documentation

4 CPs: version consistency, link validity, code example executability, EN-CN correspondence. Results: version inconsistencies 45%, CLI documentation omissions 90%.

6.3 Client Requirement Management

Case study: financial software company. Requirement changes 77% (7.3→1.7/project). First-pass acceptance 33%→100%. Rework 31%→8%. Client satisfaction 3.2→4.6/5.

6.4 New Employee Onboarding

15 CPs standardizing the process. Fitted curve: $T(n) = 5.2 \times n^{-(0.38)}$, $R^2=0.97$. By the 10th hire, onboarding takes <2.1 days. Annual ROI: 15:1.

6.5 Meeting Decision Tracking

Technical decisions no longer depend on chat history. After 6 months: accurate recall 30% (oral) → 100% (documented). Each decision follows proposal→design→archive.

7. Technical Operations Scenarios

7.1 Deployment Verification

5 CPs: image pull → container startup → health check → log anomaly → rollback readiness. Efficiency: 15–20 min → 90 sec (–92%). 30 deployments: 4 issues intercepted (13.3%).

7.2 Configuration Management

5 CPs: snapshot → validation → activation → regression → rollback. 20 changes with 0 production alerts (10% before STDD).

7.3 CI/CD Quality Gates

7 automated checks per push. Non-code projects auto-switch to document-specific dimensions (links, formats, version sync).

7.4 Incident Postmortem

The V2.7 development incident [28] generated 2 high-value experiences (EXP-2026-0005/0006) driving 5 V2.8 process fixes. Traditional postmortem = “archive”; STDD postmortem = “activate” (auto-load on next similar task).

8. Evaluation and Validation

8.1 Quantitative Results

Metric	Before	After	Improvement
Strategy omission rate	<15%	0%	–100%
Document inconsistency	2–3/version	0	–100%
Website rollback rate	15%	0%	–100%
New hire onboarding	5 days	2 days	–60%
Audit traceability	0%	100%	–100%
Experience retention	<20%	100%	+400%
Client rework rate	31%	8%	–74%
Deploy verification time	15–20min	90s	–92%

8.2 Qualitative Assessment

Operationalizability: Requirement clarity 3.2/5→4.8/5 (+50%). **Traceability:** 40+ changes with complete decision records. **Replicability:** 4/6 website CPs directly reusable. **Evolvability:** Experience curve follows Basilii’s model [12]: rapid growth (first 10 changes: 4.2/change), deceleration (11–30: 1.8/change), steady state (31+<0.5/change).

8.3 Cost-Benefit

Scenario	Investment	Annual Benefit	ROI
Quant deployment	2h	Avoids live loss	100:1+
Onboarding	4h	¥12,000	15:1
Website	30min	Avoids rollbacks	20:1
Documents	15min	Avoids errors	10:1

9. Discussion

9.1 Applicability Boundaries

STDD applies to work satisfying: (1) expected outcomes definable; (2) results objectively verifiable; (3) tasks repetitive. Not applicable to: purely creative work, one-off tasks, real-time intuitive judgment.

9.2 Limitations

Methodological: 2–3 day learning cost; 15–25 min documentation overhead per change; Git dependency limits non-technical team adoption.
Empirical: Only 3 projects, ~50 changes evaluated. Fisher exact test $p=0.24$ for website rollback data—not statistically significant. Pre-post design lacks rigorous control groups. Self-report bias affects organizational metrics. Six-month period insufficient for long-term assessment.

Technical: Agent CP requires CLI/Shell knowledge. Current 6 assertion types cannot handle image recognition, NLP, or domain expert judgment. Cross-platform compatibility (Windows/Linux/macOS) not fully verified.

Generalizability: 12 scenarios not generalizable. Healthcare, legal, education domains absent. 2–5 person teams inhibit organizational random sampling. Cultural preferences influence methodology design.

Measurement: Effect attribution faces inherent field-study challenges. ROI estimates are order-of-magnitude references.

Ethical: Balance between automation and human judgment in high-stakes decisions. Experience library may amplify initial biases.

9.3 Future Work

(1) Multi-center collaborative studies; (2) Quasi-experimental designs; (3) Domain-specific CP template libraries; (4) Extended assertion types (image, NLP); (5) Experience equality analysis; (6) Cross-cultural studies.

9.4 Relationship with Existing Frameworks

STDD is a **quality assurance layer** compatible with OpenSpec (change management), Spec Kit (requirements), and Superpowers (TDD execution).

10. Conclusion

This paper has demonstrated, through theoretical analysis and empirical validation, that STDD’s three core mechanisms are fully effective in non-programming contexts. Quantitative data from 12 scenarios demonstrate quality improvements comparable to or exceeding programming scenarios. The methodology applies to any work satisfying definable expectations, objective verifiability, and repeatability.

The essence of STDD is not a programming tool, but a methodology that transforms AI output from unpredictable to predictable—applicable to any scenario where AI can execute but quality assurance is needed.

参考文献：本文通过理论分析和实证验证，证明了 STDD 方法论可以脱离代码介体，应用于金融、日常、运维等非常规领域。Agent CP 断言是关键技术基础。12 个场景的定量数据表明质量提升显著。STDD 的本质不是编程工具，而是一种从“不确定”变为“可预测”的通用方法论。

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